

Package ‘GR2MSemiDistr’

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Type Package

Title Semi-Distributed GR2M Model for Large-Sample Hydrological Applications

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Description Implements a semi-distributed adaptation of the GR2M monthly hydrological model for large-sample and operational hydrological modeling. The model uses regional parameters and subbasin-level forcing data to simulate water balance components such as precipitation, evapotranspiration, runoff, and discharge. It supports warm-up periods, initial states, output saving, regional parameter correction, and performance evaluation through various objective functions. The routing scheme is implemented using a transfer matrix that represents subbasin connectivity from upstream to downstream. Designed for flexible integration in research, water resource assessments, and operational hydrological systems in complex basins.

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Imports airGR, hydroGOF, terra, lubridate, tictoc, sf, rtop,
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Contents

Build_Transfer_Matrix	2
Create_Forcing_Inputs	3
Load_example_data	5
Optim_GR2MSemiDistr	5
Run_GR2MSemiDistr	8
Index	12

Build_Transfer_Matrix	<i>Build transfer matrix (rows = receivers, cols = donors)</i>
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Description

Create a dense 0/1 transfer matrix (as a data.frame) from a river reach/subbasin layer that contains a unique ID ('COMID') and its immediate downstream ID ('NextDownID'). ****Rows are receivers**, **columns are donors****. The result is a square (nsub x nsub) matrix of subbasin connectivity whose row and column names must match 'COMID'.

Usage

Build_Transfer_Matrix(riv)

Arguments

riv A 'SpatVector' of river reaches/subbasins. Its attribute table must contain 'COMID' (unique subbasin ID) and 'NextDownID' (immediate downstream ID). Geometry is ignored; only attributes are used. Outlets (no downstream basin/river) must be encoded with '-1' in 'NextDownID'.

Details

Designed to plug into routing steps used by 'Run_GR2MSemiDistr()' and related functions in the package.

Value

A 'data.frame' of size n x n with entries in 0,1. Row names and column names are the node IDs (COMID domain). Outlets (no downstream) lead to rows of zeros.

Create_Forcing_Inputs *Extract and prepare model input data in the airGR format (DatesR, P, E, and optionally Q) using gridded monthly precipitation and evapotranspiration.*

Description

This function extracts spatially averaged monthly precipitation and potential evapotranspiration values for each subbasin, and optionally includes observed streamflow data, to produce a formatted data frame compatible with the airGR hydrological modeling framework.

Usage

```
Create_Forcing_Inputs(
  Subbasins,
  Precip,
  PotEvap,
  Qobs = NULL,
  DateIni,
  DateEnd,
  IniGrids = "01/1981",
  Save = FALSE,
  Update = FALSE,
  Members = NULL
)
```

Arguments

Subbasins	A SpatVector object containing the subbasins' geometries. Must include the following attributes: 'COMID' (unique identifier), and 'Region' (string code for the hydro-climatic region).
Precip	A SpatRaster object with monthly precipitation data in [mm/month].
PotEvap	A SpatRaster object with monthly potential evapotranspiration data in [mm/month].
Qobs	Optional. A vector with observed streamflow in [m ³ /s] at the basin outlet.
DateIni	Start date of the output time series in 'mm/yyyy' format.
DateEnd	End date of the output time series in 'mm/yyyy' format.
IniGrids	Initial date of the gridded precipitation and evapotranspiration datasets in mm/yyyy format.
Save	Logical. If TRUE, saves the resulting data frame to a .txt file in the './Inputs' directory. Default is FALSE.
Update	Logical. If TRUE, returns only the last month of data (useful for operational updates). Default is FALSE.
Members	Optional. Integer indicating the number of ensemble members for streamflow forecasting. If provided, the date vector will be repeated accordingly. Default is NULL.

Value

A data frame containing the model inputs in the airGR format, with columns:

DatesR Monthly dates

P_x Mean precipitation for subbasin x, where x is the COMID

E_x Mean potential evapotranspiration for subbasin x, where x is the COMID

Q Observed streamflow [m^3/s], if provided

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define simulation period and grid start date
start_date <- "01/1981"
end_date   <- "12/2016"
ini_grids  <- "01/1981"

# Create input data for the model
model_inputs <- Create_Forcing_Inputs(
  Subbasins = cat,
  Precip = grid_pr,
  PotEvap = grid_pe,
  Qobs = qobs,
  DateIni = start_date,
  DateEnd = end_date,
  IniGrids = ini_grids,
)

# Create figures to visualize the inputs
dates <- as.Date(model_inputs$DatesR)

# Plot average precipitation across subbasins
P_vars <- grep("^P_", names(model_inputs), value = TRUE)
avg_P <- rowMeans(model_inputs[, P_vars], na.rm = TRUE)
plot(dates, avg_P, type = "l", col = "blue", lwd = 1.5,
     ylab = "Precipitation [mm/month]", xlab = "Date",
     main = "Mean Precipitation across Subbasins")

# Plot average potential evapotranspiration across subbasins
E_vars <- grep("^E_", names(model_inputs), value = TRUE)
avg_E <- rowMeans(model_inputs[, E_vars], na.rm = TRUE)
plot(dates, avg_E, type = "l", col = "orange", lwd = 1.5,
     ylab = "PET [mm/month]", xlab = "Date",
     main = "Mean Potential Evapotranspiration across Subbasins")

# Plot observed streamflow at outlet (if available)
if ("Q" %in% names(model_inputs)) {
  plot(dates, model_inputs$Q, type = "l", col = "darkgreen", lwd = 1.5,
       ylab = "Q [ $\text{m}^3/\text{s}$ ]", xlab = "Date",
```

```

    main = "Observed Streamflow at Basin Outlet")
  }

```

Load_example_data	<i>Load example hydrological data included in the package</i>
-------------------	---

Description

Loads spatial and tabular example data (raster and vector) from 'inst/extdata'.

Usage

```
Load_example_data()
```

Value

A list with objects: 'cat', 'dem', 'grid_pr', 'grid_pe', 'qobs'

Examples

```

# Load example data

data <- GR2MSemiDistr::Load_example_data()
names(data)
cat    <- data$cat
dem    <- data$dem
qobs   <- data$qobs
grid_pr <- data$grid_pr
grid_pe <- data$grid_pe
matrixT <- data$matrixT

terra::plot(terra::mask(dem, cat), alpha=0.8)
terra::plot(cat, add=TRUE)
terra::text(cat)

```

Optim_GR2MSemiDistr	<i>Parameter optimization for a semi-distributed GR2M model using the SCE-UA algorithm.</i>
---------------------	---

Description

Optimizes the GR2M monthly water balance model parameters (X1, X2, fp, fe) for multiple calibration regions using the SCE-UA algorithm. The model is run in a semi-distributed setup over multiple subbasins grouped by region. Users can select from various objective functions (e.g., NSE, KGE, RMSE, and linear combinations), including composite metrics. Parameter bounds must be provided, and regions can be excluded from calibration.

Usage

```

Optim_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  MatrixTransfer,
  Outlet,
  Parameters.Min = c(100, 0.1, 0.8, 0.8),
  Parameters.Max = c(2000, 10, 1.2, 1.2),
  Optimization = "OF1",
  Max.Functions = 1000,
  WarmUp = 36,
  No.Optim = NULL,
  w1 = 0.6,
  w2 = 0.3,
  w3 = 0.2
)

```

Arguments

Data	A data frame of model input data in the airGR format, as produced by <code>Create_Forcing_Inputs</code> . It must include columns: <code>DatesR</code> , <code>P_1</code> to <code>P_n</code> , <code>E_1</code> to <code>E_n</code> , and <code>Q</code> (used for calibration).
Subbasins	A <code>SpatVector</code> object containing the geometries of subbasins. It must include attributes <code>"COMID"</code> (unique subbasin ID) and <code>"Region"</code> (region name/code).
RunIni	Simulation start date in the format <code>"mm/yyyy"</code> .
RunEnd	Simulation end date in the format <code>"mm/yyyy"</code> .
Parameters	A data frame containing GR2M model parameters and correction factors per region. Must have columns: <code>Region</code> , <code>X1</code> , <code>X2</code> , <code>fp</code> , <code>fe</code> .
MatrixTransfer	Square matrix or data frame (<code>nsub</code> × <code>nsub</code>) of subbasin connectivity. Row and column names must match <code>COMID</code> and are internally reordered to align with P/E inputs and QS.
Outlet	Optional. Outlet subbasin identifier as a <code>COMID</code> code present in <code>Subbasins</code> . If provided, the routed outlet series is extracted from this column.
Parameters.Min	Numeric vector of length 4, specifying lower bounds for optimization in the following order: <code>c(X1, X2, fp, fe)</code> .
Parameters.Max	Numeric vector of length 4, specifying upper bounds for optimization in the same order as <code>'Parameters.Min'</code> .
Optimization	Character. Objective function to optimize. Available options include: "OF1" Kling-Gupta Efficiency (KGE). "OF2" Nash-Sutcliffe Efficiency (NSE). "OF3" NSE applied to log-transformed flows (log-NSE, Pushpalatha 2012). "OF4" Root Mean Square Error (RMSE). "OF5" Percent Bias (PBIAS). "OF6" FDC Bias (FDC Bias), based on flow duration curves. "OF7" Pearson correlation coefficient (<code>r</code>).

	"OF8" Composite objective: $w1 \times KGE + w2 \times NSE - w3 \times RMSE$.
	"OF9" Composite objective: $w1 \times KGE + w2 \times \log\text{-}NSE - w3 \times RMSE$.
	"OF10" Composite objective: $w1 \times KGE.km + w2 \times KGE.lf - w3 \times RMSE$.
Max.Functions	Integer. Maximum number of function evaluations in the optimization. Default is 1500.
WarmUp	Optional number of months to discard from the beginning of the simulation as warm-up. Default is NULL.
No.Optim	Character vector (optional). Region names to exclude from the optimization (parameters will be kept fixed).
w1	Numeric. Weight for the first component in composite objective functions (OF8, OF9, OF10). Default is 0.6.
w2	Numeric. Weight for the second component in composite objective functions. Default is 0.3.
w3	Numeric. Weight for the third component in composite objective functions. Default is 0.2.

Value

A list with the following elements:

Parameters Data frame of optimized parameters per region (columns: Region, X1, X2, fp, fe).

OF Final value of the selected objective function.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define initial parameters for each region
param_init <- data.frame(
  Region = unique(cat$Region),
  X1 = rep(500, length(unique(cat$Region))),
  X2 = rep(1.5, length(unique(cat$Region))),
  fp = rep(1.0, length(unique(cat$Region))),
  fe = rep(1.0, length(unique(cat$Region)))
)

# Optimize parameters using the KGE criterion (OF1)
result <- Optim_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  MatrixTransfer=matrixT,
  Outlet = '8', # basin outlet comid
  Parameters = param_init,
  Optimization = "OF10"
)
```

```

# Extract results
best_params <- result$Parameters
final_score <- result$OF

# Plot observed vs simulated discharge at the outlet
model <- Run_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  MatrixTransfer=matrixT,
  Outlet = '8', # basin outlet comid
  Parameters = best_params,
)

if (!is.null(model$SINK)) {
  par(mfrow = c(1, 1))
  plot(dates, model$SINK$sim, type = "l", col = "blue",
       xlab = "Date", ylab = "Discharge [m³/s]",
       lines(dates, model$SINK$obs, col = "red", type='o'))
  legend("topright", legend = c("Simulated", "Observed"),
       col = c("blue", "red"), lty = c(1, 2), lwd = 2, bty = "n")
}

```

Run_GR2MSemiDistr	<i>Run the GR2M model for multiple subbasins (semi-distributed with routing)</i>
-------------------	--

Description

This function executes the monthly GR2M water balance model across multiple subbasins using a semi-distributed approach. It applies region-specific parameters and correction factors for precipitation and potential evapotranspiration, routes the simulated flows through the river network via a transfer matrix, and can optionally save results as text files.

Usage

```

Run_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  MatrixTransfer,
  Outlet = NULL,
  WarmUp = NULL,
  IniState = NULL,
  Save = FALSE,
  Update = FALSE
)

```

Arguments

Data	A data frame of model input data in the airGR format, as produced by Create_Forcing_Inputs. It must include columns: DatesR, P_1 to P_n, E_1 to E_n, and Q (used for calibration).
Subbasins	A SpatVector object containing the geometries of subbasins. It must include attributes "COMID" (unique subbasin ID) and "Region" (region name/code).
RunIni	Simulation start date in the format "mm/yyyy".
RunEnd	Simulation end date in the format "mm/yyyy".
Parameters	A data frame containing GR2M model parameters and correction factors per region. Must have columns: Region, X1, X2, fp, fe.
MatrixTransfer	Square matrix or data frame (nsub $\times$ nsub) of subbasin connectivity. Row and column names must match COMID and are internally reordered to align with P/E inputs and QS.
Outlet	Optional. Outlet subbasin identifier as a COMID code present in Subbasins. If provided, the routed outlet series is extracted from this column.
WarmUp	Optional number of months to discard from the beginning of the simulation as warm-up. Default is NULL.
IniState	Optional list of initial states for each subbasin. Default is NULL.
Save	Logical. If TRUE, output time series are saved as tab-separated text files under "Outputs/".
Update	Logical. If TRUE and Save = TRUE, removes previous month's output files before saving the current ones.

Details

Subbasin-level simulated discharge (in m³/s) is obtained from GR2M runoff (mm/month) using basin area and days per month, and then accumulated/routed through the network using the geometric series of the transfer matrix $S = I + T + T^2 + \dots + T^{(n-1)}$, applied as $QR = QS * S'$. The outlet series is taken from the routed matrix column corresponding to the provided outlet COMID.

Value

A list with:

PR Matrix of total precipitation [mm/month] per subbasin.

AE Matrix of actual evapotranspiration [mm/month] per subbasin.

SM Matrix of production store level [mm/month] per subbasin.

PC Matrix of percolation [mm/month] per subbasin.

RU Matrix of runoff [mm/month] per subbasin (model internal output).

QR Matrix of routed discharge [m³/s] per subbasin after network accumulation using MatrixTransfer.

Dates Date vector for the simulation period.

COMID Vector of subbasin COMIDs (column order matches matrices).

EndState List of final state variables for each subbasin.

SINK Only if outlet is provided. Data frame with outlet series where sim is the routed outlet discharge (column QR[, Outlet]). If observed outlet discharge QR exists in Data, obs is included as well. Row names are Dates.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define region parameters
param_df <- data.frame(
  Region = unique(cat$Region),
  X1 = 2000,
  X2 = 1.8,
  fp = 0.8,
  fe = 0.8
)

# Run the semi-distributed GR2M model
model <- Run_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2016",
  MatrixTransfer = matrixT,
  Outlet = '8', # basin outlet comid
  Parameters = param_df,
)

# Select COMID of the subbasin to plot
target_comid <- model$COMID[1] # Change index as needed
idx <- which(model$COMID == target_comid)
dates <- as.Date(model$Dates)

# Plot model outputs for selected subbasin
par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))

plot(dates, model$PR[, idx], type = "l", col = "blue",
     main = paste("Precipitation (PR) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$AE[, idx], type = "l", col = "orange",
     main = paste("Actual Evapotranspiration (AE) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$SM[, idx], type = "l", col = "green4",
     main = paste("Production Store (SM) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$PC[, idx], type = "l", col = "purple",
     main = paste("Percolation (PC) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")

plot(dates, model$RU[, idx], type = "l", col = "darkred",
     main = paste("Runoff (RU) - COMID", target_comid),
     xlab = "Date", ylab = "mm/month")
```

```
plot(dates, model$QR[, idx], type = "l", col = "blue",
     main = paste("Simulated Discharge (QR) - COMID", target_comid),
     xlab = "Date", ylab = "m3/s")

# Optional: compare simulated vs observed outlet discharge
if (!is.null(model$SINK)) {
  par(mfrow = c(1, 1))
  plot(dates, model$SINK$sim, type = "l", col = "blue",
       xlab = "Date", ylab = "Discharge [m3/s]",
       lines(dates, model$SINK$obs, col = "red", type='o'))
  legend("topright", legend = c("Simulated", "Observed"),
        col = c("blue", "red"), lty = c(1, 2), lwd = 2, bty = "n")
}
```

Index

Build_Transfer_Matrix, [2](#)

Create_Forcing_Inputs, [3](#)

Load_example_data, [5](#)

Optim_GR2MSemiDistr, [5](#)

Run_GR2MSemiDistr, [8](#)